

3D Modeling

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Introduction

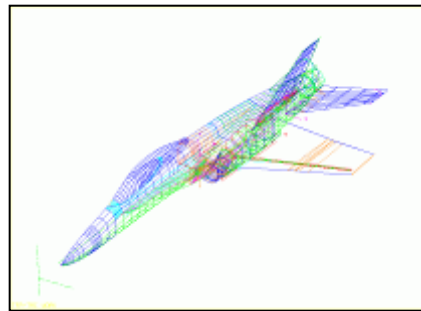
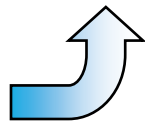
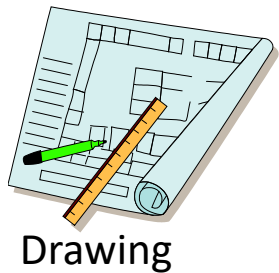
Introduction

- How to model real world objects? - Design
 - How to put forth ideas in visual manner – Communication
 - How to verify that design serves the purpose – Analysis
 - How to get it made? – Manufacturing
 - All the above can happen without Computers. But
 - Better if assisted by Computers/Software
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- That's why : Computer Aided < > (CAx)

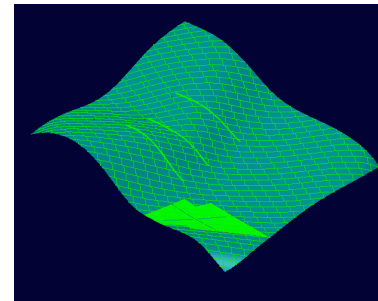
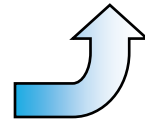
History

- The first source of CAD resulted from attempts to automate the drafting process.
- These developments were pioneered by the General Motors Research Laboratories in the early 1960s.
- CAD became more widely used after 1970 because of technological advancements.
- CAD allowed users to design products much quicker without the production of an actual product.

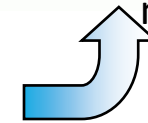
Evolution of CAD Technology



Wire frame
model



Surface
model



Solid
model

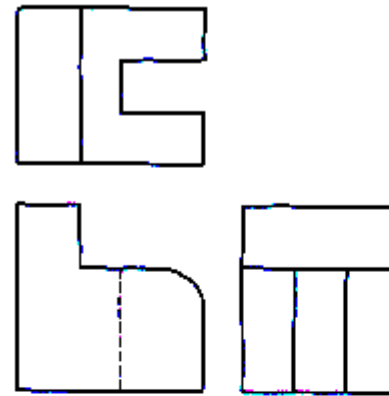


Manual drafting

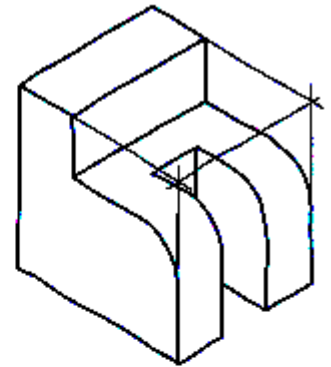
Since 1970's: electronic drafting board

Manual Drafting

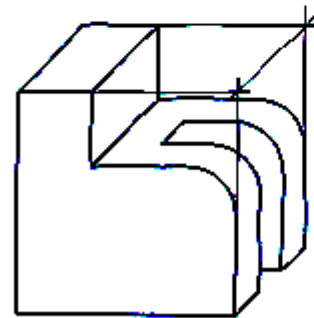
- 2D representations used to represent 3D objects
 - multi-view drawings
 - pictorials
- Standards and conventions developed so that 3D object could be built from drawings
- Drawings created manually or using 2D CAD
- Difficult to visualize, error-prone, time-consuming



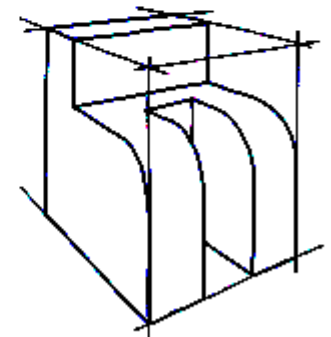
(A) Multiview



(B) Axonometric



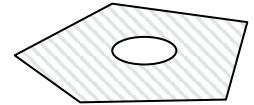
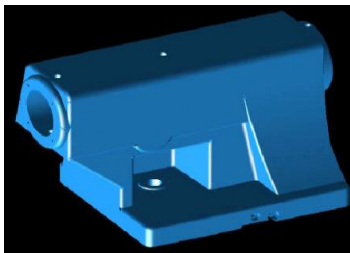
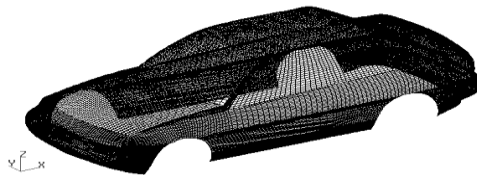
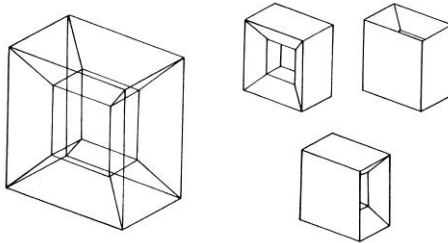
(C) Oblique



(D) Perspective

CAD - Types

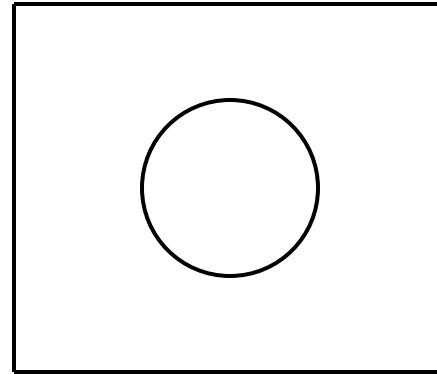
- 2D model: Point, line, circular arc, planar curve
- 3D model
 - Wire frame
 - Surface
 - Solid

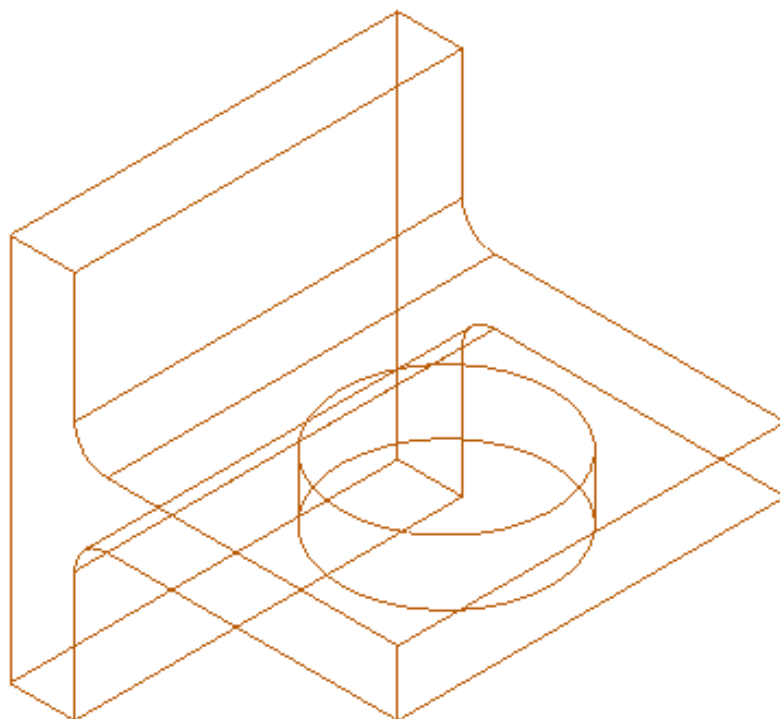


Advantages and Disadvantages of each?

2D CAD

- Simply replaces manual drawing
- Provides a set of drawing tools to create 2D elements
 - Lines, circles, arcs, etc.
- More accurate, easier changes to drawings
- Still no 3D representation of the object
- Example: AutoCAD

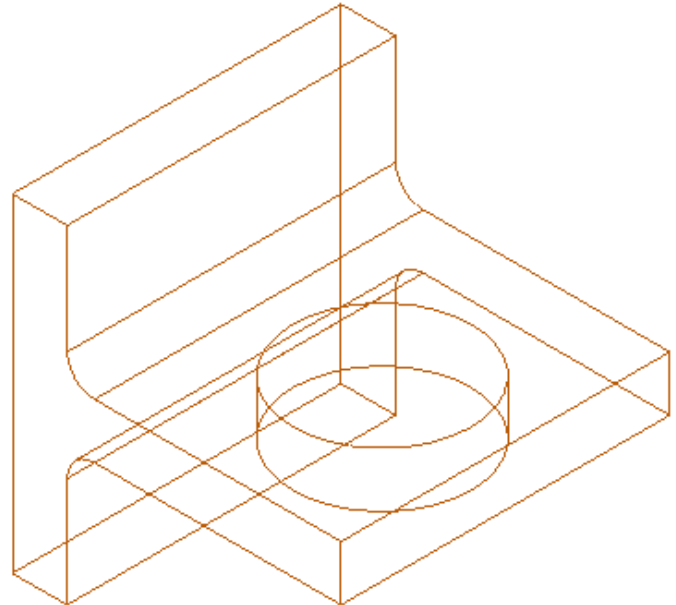




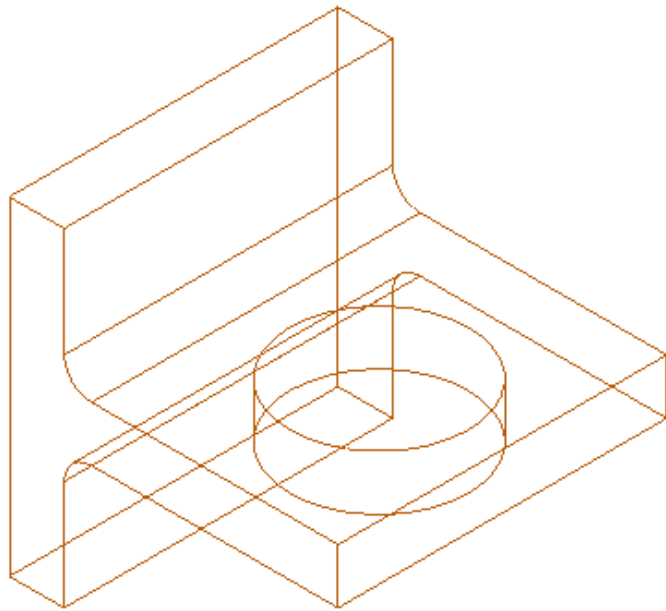
Early 1980's: wire frame geometry

3D Wire frame Modeling

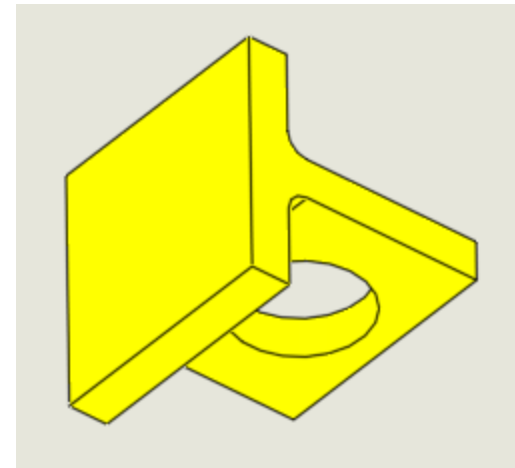
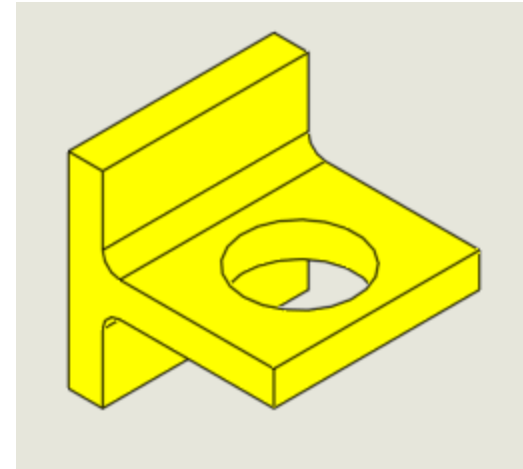
- Geometric entities are lines and curves in 3D
- Volume or surfaces of object not defined
- Easy to store and display
- Hard to interpret - ambiguous

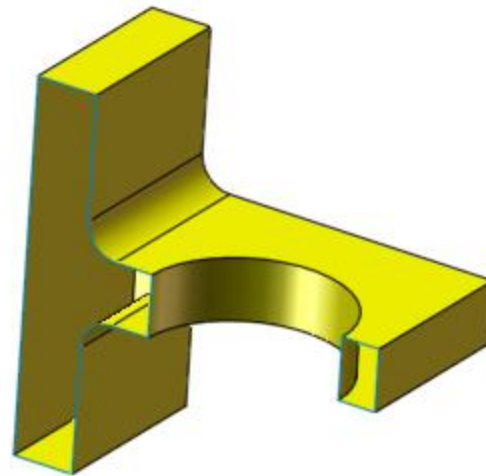
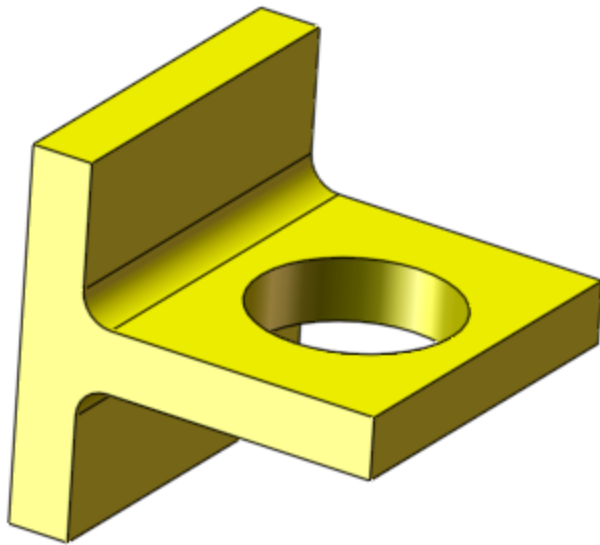


What is this?



Problems with wire frame models

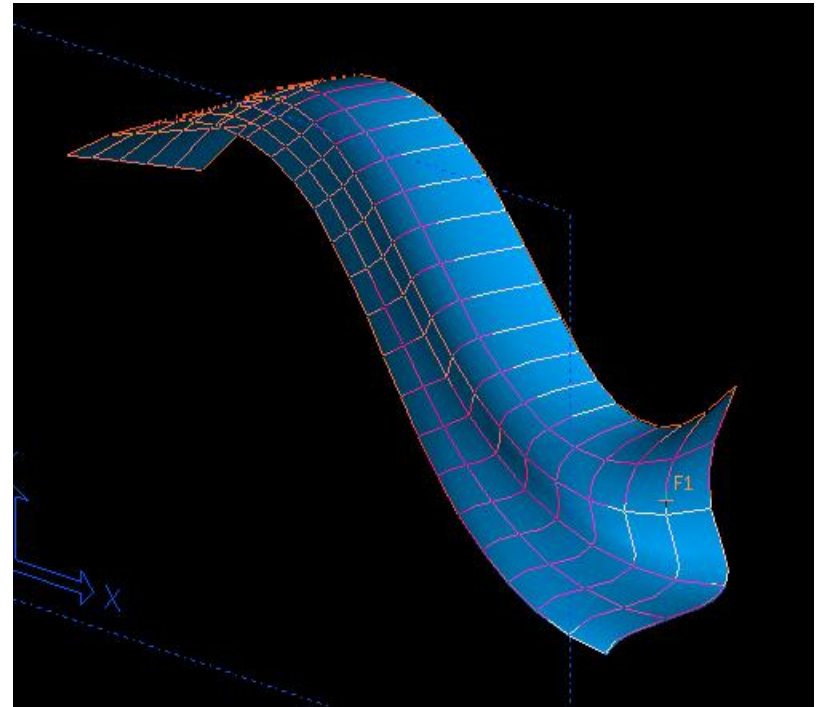




Late 1980's: Surface Modeling

3D Surface Modeling

- Models 2D surfaces in 3D space
- All points on surface are defined
 - useful for machining, visualization, etc.
- Surfaces have no thickness, objects have no volume or solid properties
- Surfaces may be open



A Surface Model created using Alias StudioTools



Surface Model created using Rhino

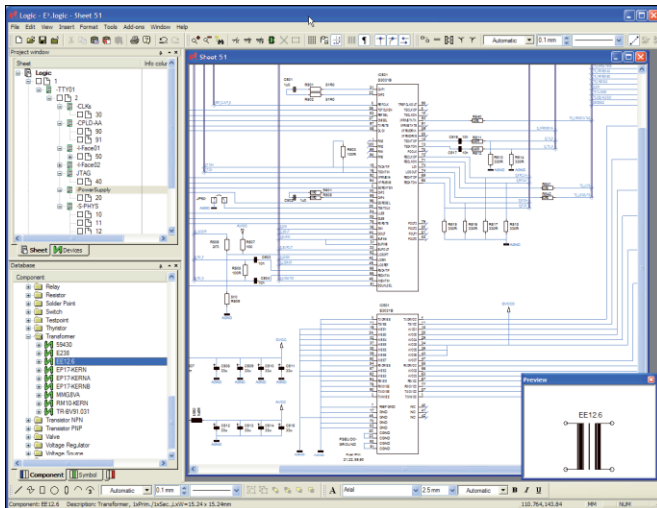


Why draw 3D Models?

- 3D models are easier to interpret.
- Less expensive than building a physical model.
- 3D models can be altered easily, create more concepts.
- 3D models can be used to perform engineering analysis, finite element analysis (stress, deflection, thermal.....) and motion analysis.
- 3D models can be used directly in manufacturing, Computer Numerical Control (CNC).

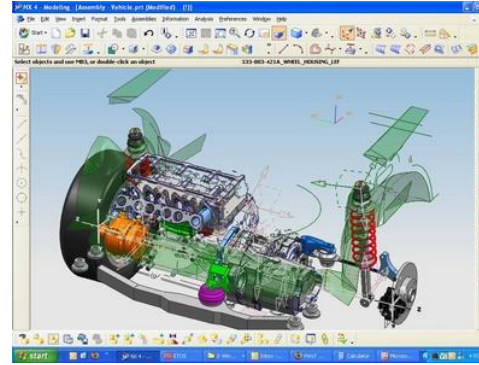
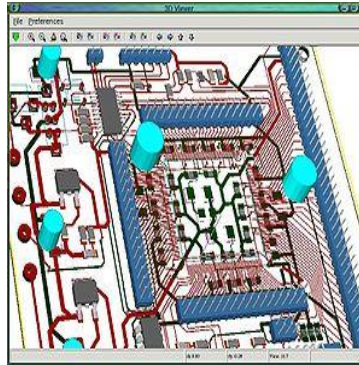
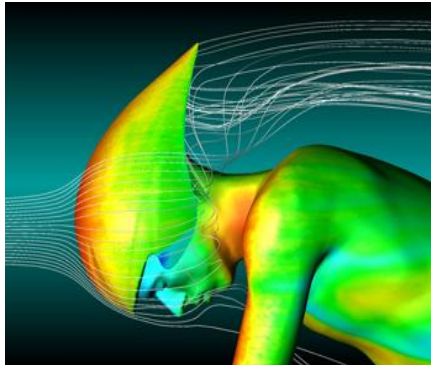
2D Applications

- Drafting – sketches, architectures, Drawings
- Art – Sketches, painting
- Electronic layouts, circuit design



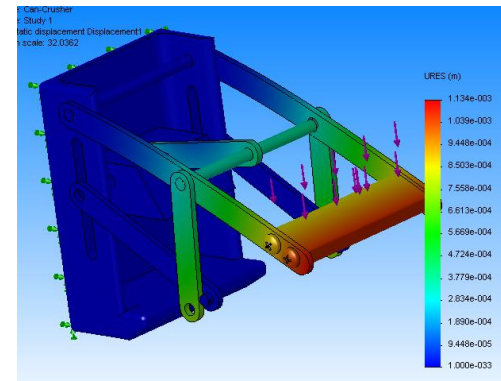
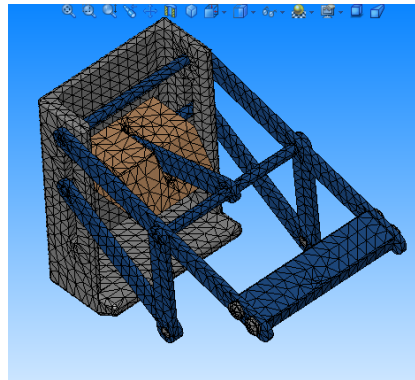
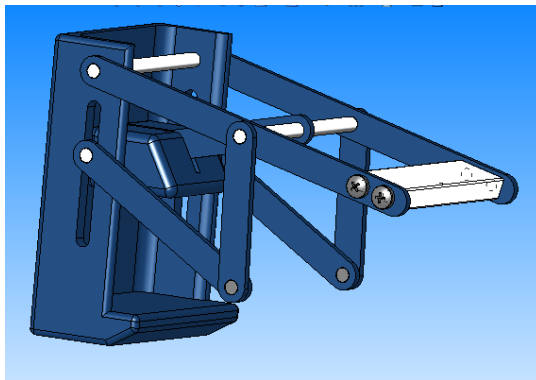
3D Applications

- CAD (Computer Aided Design)
- CAM (Computer Aided Manufacturing)
- CAE (Computer Aided Engineering) Finite Element Method
- CG (Computer Graphics)



Basics of Finite Element Analysis (FEA)

- A complex problem is divided into a smaller and simpler problems that can be solved by using the existing knowledge of mechanics of materials and mathematical tools
- Modern mechanical design involves complicated shapes, sometimes made of different materials that as a whole cannot be solved by existing mathematical tools. Engineers need the FEA to evaluate their designs



Computer Numerical Control (CNC)

A CNC machine is an NC machine with the added feature of an on-board computer.



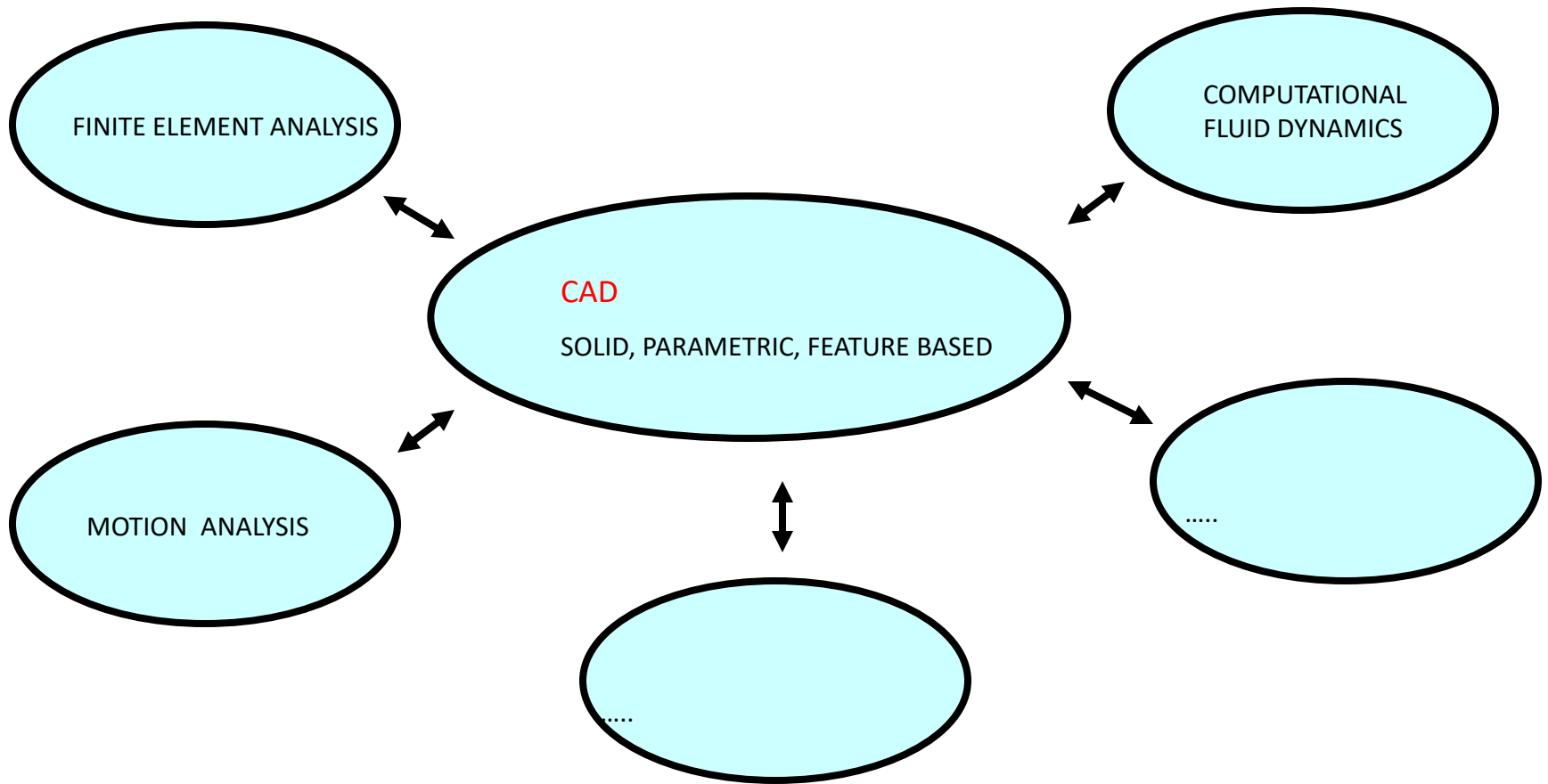
Modelling

Solid, parametric, feature based modeling

- Complete and unambiguous
- **Solid** - models have volume, and mass properties
- **Feature based** - geometry built up by adding and subtracting features
- **Parametric** - geometry can be modified by changing dimensions



MODERN CAE TOOLS



CAD (Computer Aided Design) is at the hub of other CAE (Computer Aided Engineering) tools

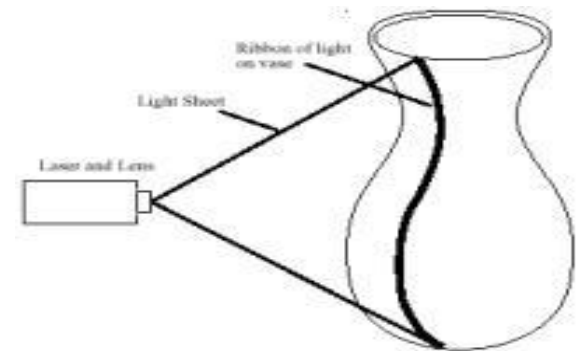
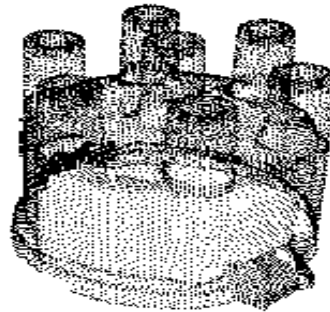
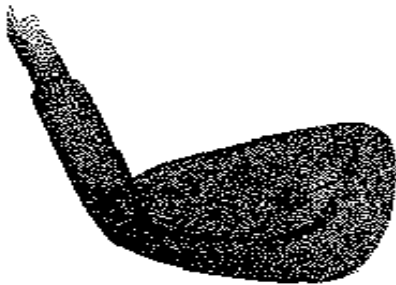
Solids

What is Solid?

- Define Solid?
- How would you represent Solid in software (data model)?

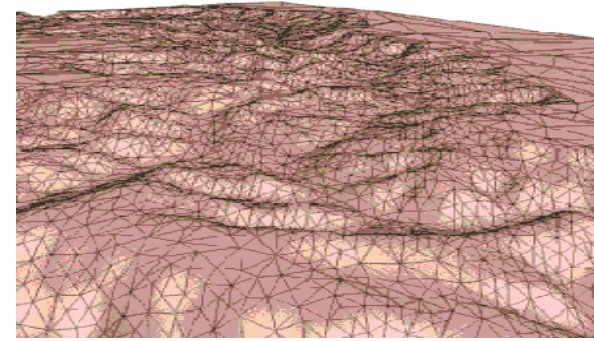
Cloud of points

- The simplest form
- Unorganized / organized points
- Too many points to represent the desired shape
- Hard to handle → further processing is required
- Obtained by digitizing
 - CMM (coordinate measuring machine)
 - Laser range scanner
 - ...



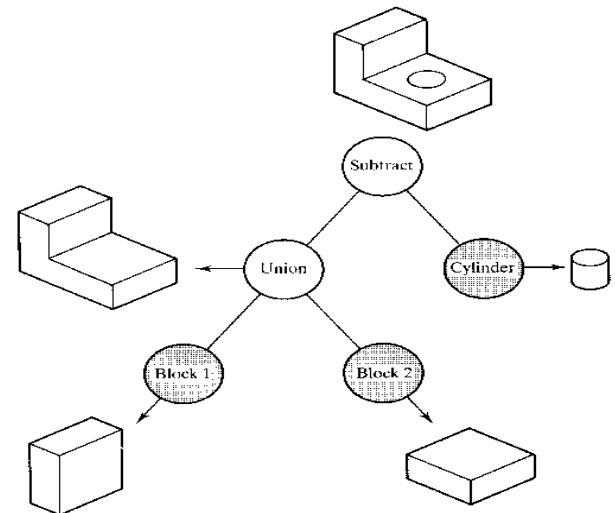
Mesh

- Most popular approximation model
- Graphics, RP, CAD/CAM, DMU, CAE
- Hard to handle
- **Triangular mesh**, Quad mesh, General polygonal mesh
- Create mesh by
 - **triangulating** cloud of points
 - **faceting** exact surface model
- Example: 123D Catch



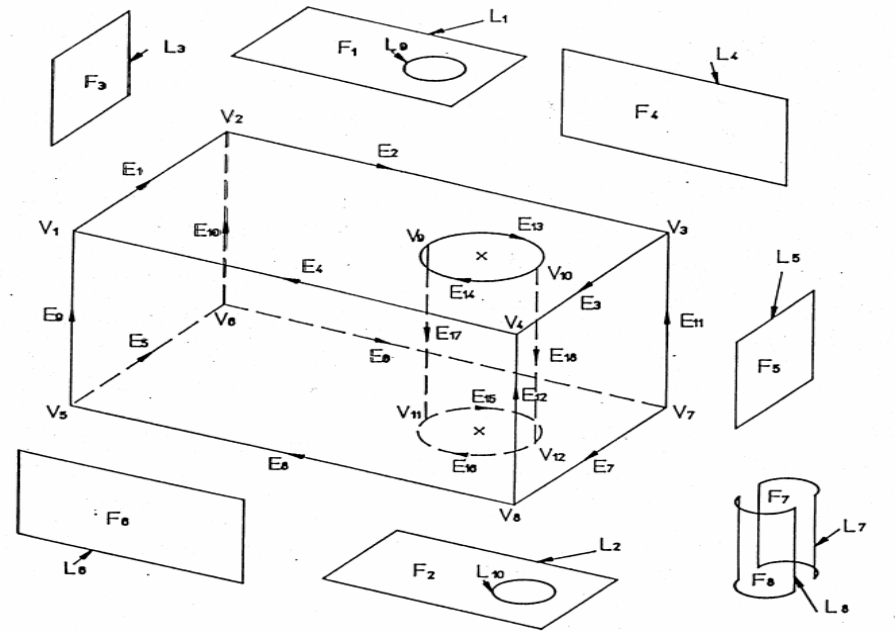
What to store : Modeling procedure

- Procedural model: CSG (Constructive Solid Geometry)
- Primitive solids with Boolean operation



What to store : result

- B-Rep (Boundary representation) model : Modeling using bounding surfaces
- Topology : connectivity
- Geometry : shape



B-Rep model

- Topological element

- Vertex
- Edge
- Loop (Edge list)
- Face
- Lump
- Body

- Geometrical element

- Point
- Curve
- Composite curve
- Surface, trimmed surface
- N/A
- N/A

Euler-Poincare formula:

For a polyhedron

$$V - E + F - 2 = 0$$

- V = Vertices
- E = Edges
- F = Faces

Example: A tetrahedron has four vertices, four faces,
and six edges

$$4 - 6 + 4 = 2.$$

Extension of solids

A solid can have holes

A face may have a loop or ring of vertices `floating', i.e.
unconnected by edges to the other vertices of the face

Extension of Euler-Poincare formula to 2-manifolds

$$V-E+F-H=2(C-G)$$

- V = Vertices
- E = Edges
- F = Faces
- H = Holes in faces
- C = Components (or shells)
- G = Genus (holes through solid)

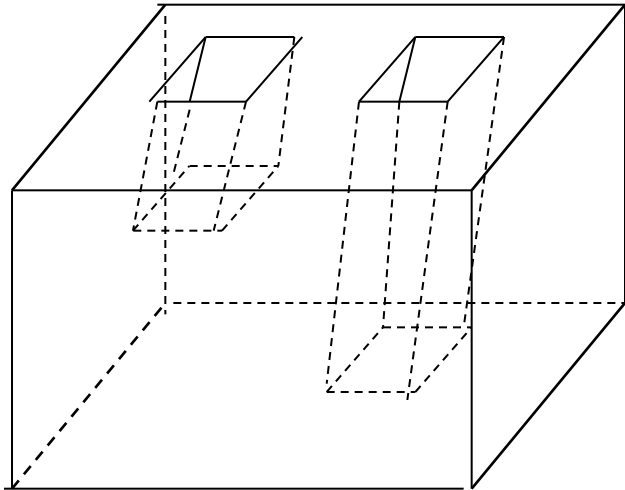
“Tweaking” (deformations, twistings, and stretchings but not tearing, or cutting) solids modifies the solid without changing the topology or the above numbers.

A solid with holes and loops

Example

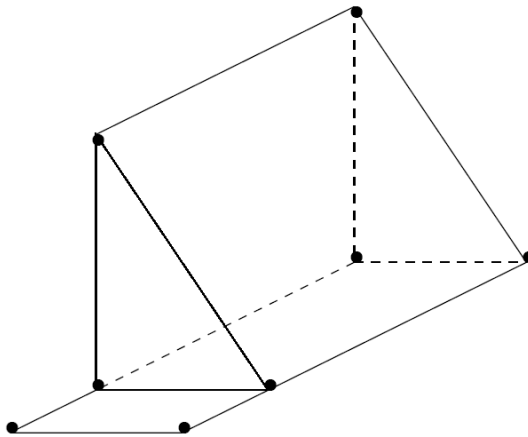
$$V - E + F - H = 2(C - G)$$

$$24 - 36 + 15 - 3 = 2(1 - 1)$$



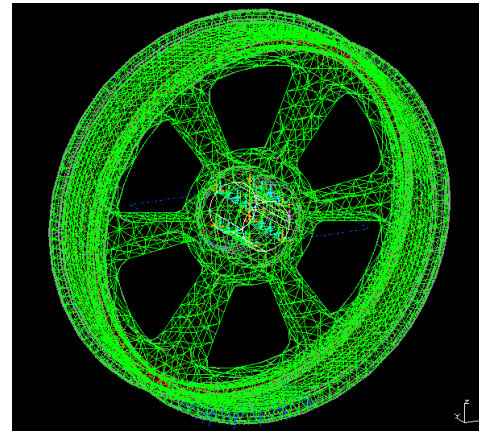
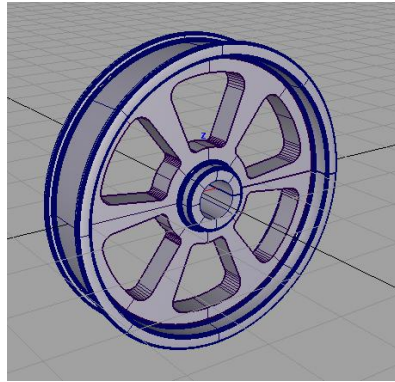
Euler Poincare' Formula

- Necessary but not sufficient condition for a valid representation.
- Example: 8 vertices, 12 edges, 6 faces

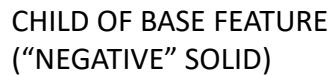


Brep vs Mesh (Design Desktop vs Catch)

- The object is represented by subdivision/discretization such as mesh and other geometric primitives.



Parametric, Feature-based Solid Model



Solid, parametric, feature-based Modeling Software

- High-end (more powerful)
 - NX (UGS)
 - Catia (Dassault Systèmes)
 - Pro/Engineer (Parametric Technologies Corp.)
- Mid-Range (easier to use)
 - Solid Edge (UGS)
 - Inventor (Autodesk)
 - SolidWorks (SolidWorks Corp.)

They all work basically the same way

References

- Ken Youssefi, “Introduction to Solid Modeling”
- Texas A & M, “Design Intent and Modeling Tools”
- Paul Kurowski, ‘Computer Aided Design (CAD)’